

ELEC4631: Review Lecture

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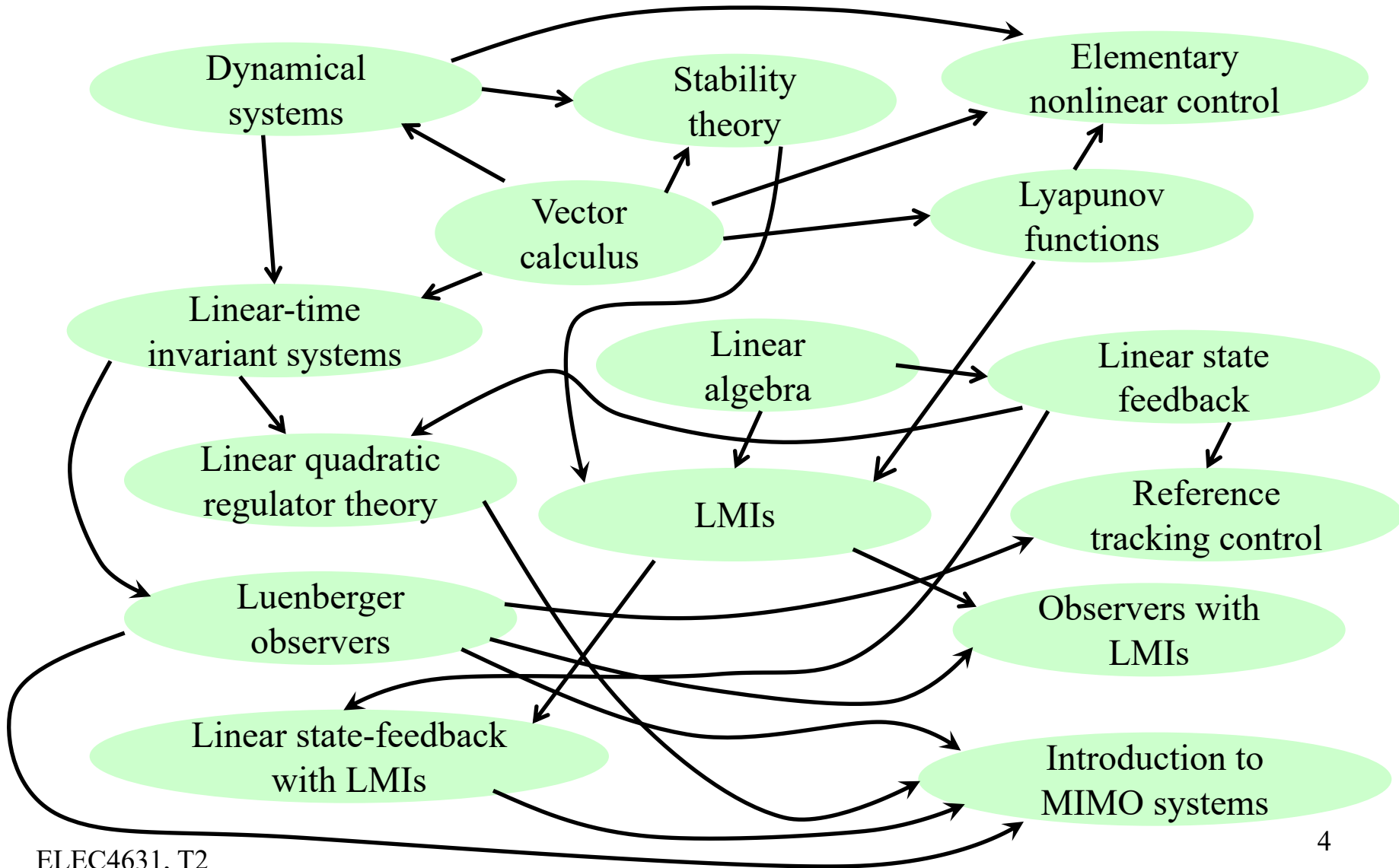
Today

- Review of materials covered in the term
- Information about final exam

Key topics

- The course has introduced many important concepts and methods in systems and control:
 - Stability and Lyapunov stability analysis
 - Nonlinear and linear state-space models
 - Controllability and observability, canonical forms
 - Controller design methods: Lyapunov, pole placement, LQR, tracking controllers;
 - Luenberger observers
 - Modern numerical tools – LMIs

“Mind map”



Part I:

Materials before flexibility week

1. Review of linear algebra and vector calculus

- Eigenvalues and eigenvectors.
- Diagonalisable matrices and diagonalisation.
- Positive definite, positive semidefinite, negative definite, negative semidefinite symmetric matrices. Characterisation.
- Change of basis. Similarity transformation.
- Jordan canonical form.

1. Review of linear algebra and and vector calculus

- Quadratic forms. Positive definite, positive semidefinite, negative definite, negative semidefinite quadratic forms. Characterisation.
- Exponentiation of a matrix.
- Vector calculus: scalar fields, level sets, gradients.

2. Dynamical systems and linear state-space systems

- State-space, states, continuous-time dynamical systems represented by ODEs, vector fields
- General SISO linear state-space systems (possibly linear time-varying).
- Linearity property with respect to the quartet of input, state, output, and initial condition.
- Time-invariance property
- SISO linear time-invariant (LTI) state-space systems.

2. Dynamical systems and linear state-space systems

- Transfer functions
- State-space representations of SISO LTI systems represented by ODEs and transfer functions: control canonical form, observer canonical form, and modal canonical form.
- Solution of SISO LTI systems

3. Lyapunov stability analysis

- Equilibrium points of ODEs.
- Stability of equilibrium points: stability in the sense of Lyapunov (i.s.L), asymptotic stability, global asymptotic stability.
- Lyapunov functions.
- Second/direct method of Lyapunov

4. Stability, controllability and observability for LTI systems

- Determining e^{Jt} for a Jordan block J using matrix Laplace transforms.
- Internal stability: stable (asymptotically stable, and marginally stable) systems, unstable systems.
- Characterisation of asymptotically stable, marginally stable and unstable systems via eigenvalues of A matrix.

4. Stability, controllability and observability for LTI systems

- BIBO stability. Criteria for BIBO stability.
- Notion of controllability. Kalman's characterisation via controllability matrix $\mathcal{C}$.
- Notion of observability. Kalman's characterisation via observability matrix $\mathcal{O}$.
- Uncontrollable and unobservable systems. Relation to pole-zero cancellation in transfer functions.

5. State-space transformations and pole placement

- Similarity transformations on SISO LTI state-space systems.
- System controllable $\leftrightarrow$ similarity transformation to control canonical form.
- System observable $\leftrightarrow$ similarity transformation to observer canonical form.
- Similarity transformation to modal canonical form.

5. State-space transformations and pole placement

- Linear state-feedback for systems in control canonical form: can place closed-loop poles anywhere.
- Generalisation to controllable systems: can place closed-loop poles anywhere.
- Pole placement using Ackermann's formula.

6. Linear quadratic regulator (LQR) theory

- Trade-off between control performance and control effort.
- Linear quadratic performance index – finite horizon and infinite horizon LQR.
- Finite horizon LQR: Riccati differential equation (RDE), optimal linear time-varying state feedback, optimal finite horizon performance index.

6. Linear quadratic regulator (LQR) theory

- Infinite horizon LQR as limit of finite horizon LQR. Conditions for solvability.
- Infinite horizon LQR: continuous-time algebraic Riccati equation (CARE), optimal linear time-invariant state feedback, optimal infinite horizon performance index.
- Stabilising property of infinite horizon LQR.

Part II:

Materials after flexibility week

7. Observers, output feedback control and the separation principle.

- State observers: estimating the state $x(t)$ through the output signal $y(t)$.
- Luenberger observer for SISO LTI state-space systems.

7. Observers, output feedback control and the separation principle.

- Observability and pole placement in the Luenberger observer.
- Observer gain design with Ackermann's formula for observer.
- Output feedback control: combining state feedback with observer.
- Separation principle for linear state-feedback and Luenberger observer.

8. Stability analysis with first method of Lyapunov

- Stability analysis of systems represented by ODEs with first/indirect method of Lyapunov.
- Checking stability of equilibrium point by examining eigenvalues of a Jacobian matrix.

9. Introduction to LMIs

- Introduction to convex sets.
- Basic definition of LMIs. LMIs in standard form.
- Convexity of LMI solutions when LMI is feasible.
- Affine maps of matrix variables. LMIs in matrix variables. Relation to LMIs in standard form.

9. Introduction to LMIs

- Schur complement of a matrix.
- Checking convexity and finding solutions of hard nonlinear inequalities via Schur complement and LMIs.

10. Controller and observer synthesis via LMIs.

- Convergence rate of a SISO LTI system.
- Verifying convergence rate as an LMI feasibility problem.
- Stabilising linear state-feedback via LMIs. Change of variable.
- Stabilising linear state feedback with convergence rate specification via LMIs.

10. Controller and observer synthesis via LMIs.

- Luenberger observers via LMIs.
- Luenberger observer with observer convergence rate specification via LMIs.

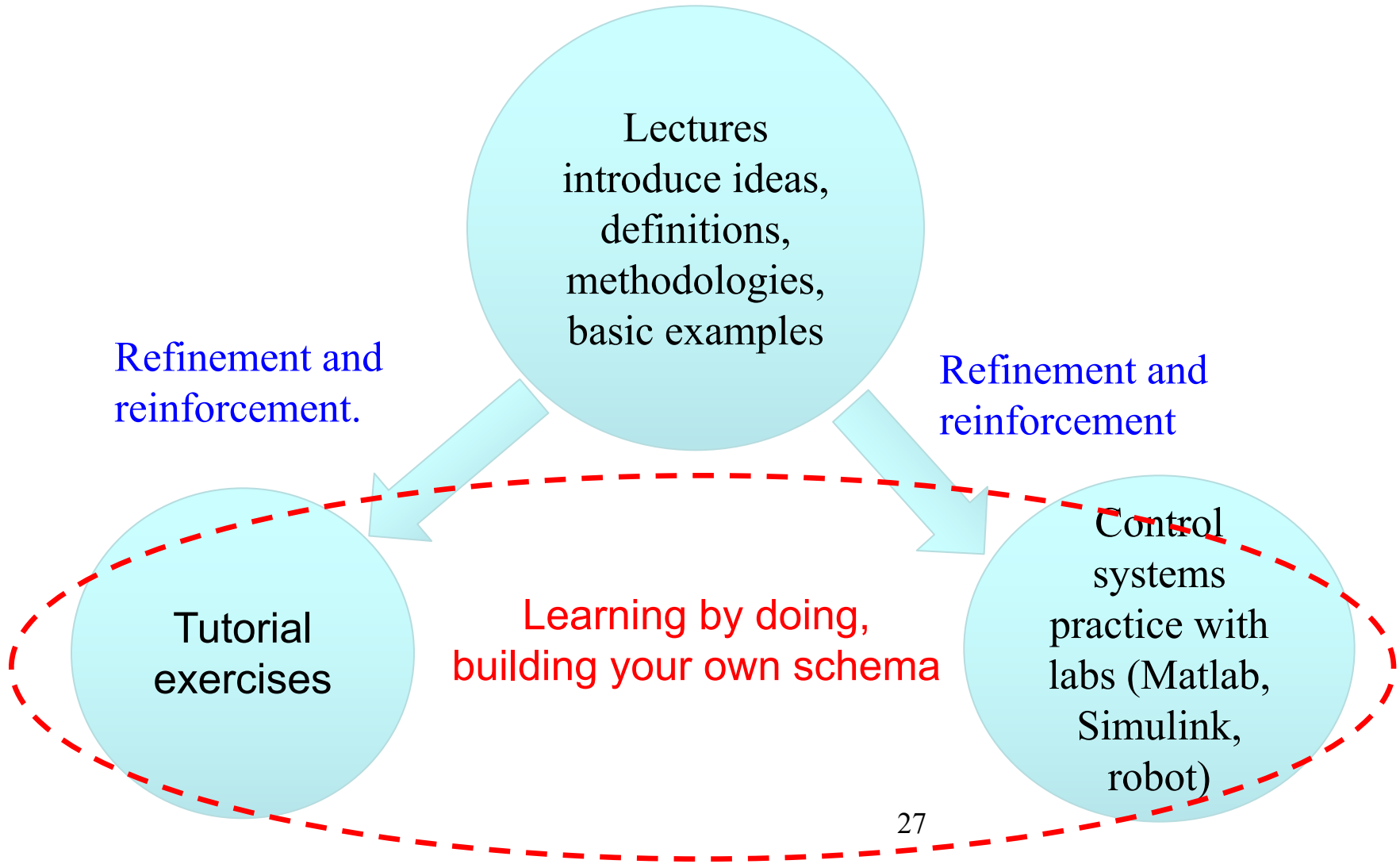
11. Reference tracking with state and output feedback

- Step input tracking with linear state-feedback.
- Step input tracking with output feedback via an output feedback controller with a zero assignment estimator. Truxal's formula. Design principles.
- Separation principle for output feedback controller with a zero assignment estimator.

12. State-feedback control design using Lyapunov functions

- State-feedback for dynamical systems governed by ODEs.
- Extension of Lyapunov functions for state-feedback control design.
- Illustrative examples of Lyapunov based state-feedback design: motion control problems and fully actuated mechanical systems.

Integrated course components



Summary

- This course has equipped you with a breadth of materials related to modern control theory.
- It has also provided you with some background for further self-study of advanced control theory or to take more advance courses.

Where to go from here?

- If you have a strong interest in studying more control, other courses that are available:
 - ELEC4632 Computer Control Systems
 - ELEC9731 Linear and Robust Control Systems
 - ELEC9732 Analysis and Design of Nonlinear Control
 - ELEC9782 Special Topics in Electrical Engineering 2 - Quantum Control

ELEC9782 Quantum Control

- This postgraduate level course is being offered in T3 2023, as a special topics course.
- Coverage is control for systems that are governed by the laws of quantum mechanics.
- This course requires existing knowledge of quantum mechanics from ELEC3705 Fundamentals of Quantum Engineering or a physics course. Should not be attempted otherwise.

Quiz 2, lab marks and final consultations

- Quiz 2 marks will be released this week
- Lab marks will be released early next week (Week 11)
- Final consultation times for the course will be this week (Week 10) Monday and Tuesday afternoons.

Final exam

- Exam duration: 2 hours
- Reading time: 10 minutes
- 4 questions containing subproblems
- Worth 56% of your total marks

Final exam

- Closed-book exam. But you may bring one A4 paper with notes on one side only.
- You are allowed to bring a UNSW approved calculator (with the UNSW approved sticker on it).
- Exam can include any materials from Weeks 1 to 10. Materials in later weeks are tied to materials from earlier weeks.

Final exam

- A list of formulas will be provided on exam paper (more on this later).
- Put your name and uni ID on both your exam paper and answer booklets.
- Exam paper to be returned with your answer booklets.

Final exam

- Bring pens for writing down your solution and pencils and a ruler for any sketches that you may be required to make.
- Final answer must be written in pen, trace over any sketches with a pen.
- **You must show all workings in your answers. Do not leave the workings on the scratch section of the exam booklets or on the exam paper; no workings, no marks.**

Suggestions

- Use the reading time to carefully read the questions. Solve the problems that are being asked.
- Problems are not grouped according to difficulty but to common themes. A challenging problem can appear in any question.
- Pace yourself. Keep track of the time.

Exam preparation

- For the final exam, I expect you to:
 - Have thoroughly revised all course materials
 - Have understood how solve the various types of problems discussed in-class, all tutorial problem sets
 - Be able to solve variations or extensions of these problems
- A practice final exam paper and some past final exam papers will be posted. Use this as an additional exercise to test your mastery of the course materials.
- However, only doing the practice paper is NOT a substitute for a proper preparation towards the final exam.

List of formulas

- The formulas below in blue will be provided. Exact list will be posted on Moodle.
 - Controllability matrix.
 - Observability matrix.
 - Transformation matrix to control canonical form.

List of formulas

- Transformation matrix to observer canonical form.
- Solution of state-space systems.
- e^{Jt} for a Jordan block J .
- Ackermann's formula for feedback gain K .
- Riccati differential equation (RDE).

List of formulas

- Optimal finite horizon LQR control signal, optimal finite horizon LQR performance index.
- Continuous algebraic Riccati equation (CARE).
- Optimal infinite horizon LQR control signal, optimal infinite horizon LQR performance index.

List of formulas

- Observer state-space equations.
- Schur complement of a symmetric matrix.
- LMI for design of feedback gain with minimum convergence rate specification. Formula for K .
- LMI for design of observer gain with minimum convergence rate specification. Formula for L .

List of formulas

- State-feedback for step input tracking.
- Polynomial for the zeros of a SISO LTI system.
- Output feedback controller with zero assignment estimator: formula for $\gamma(s)$, formula for DC gain, closed-loop zeros, formula for closed-loop characteristic polynomial, Truxal's formula.